NicheExplorer

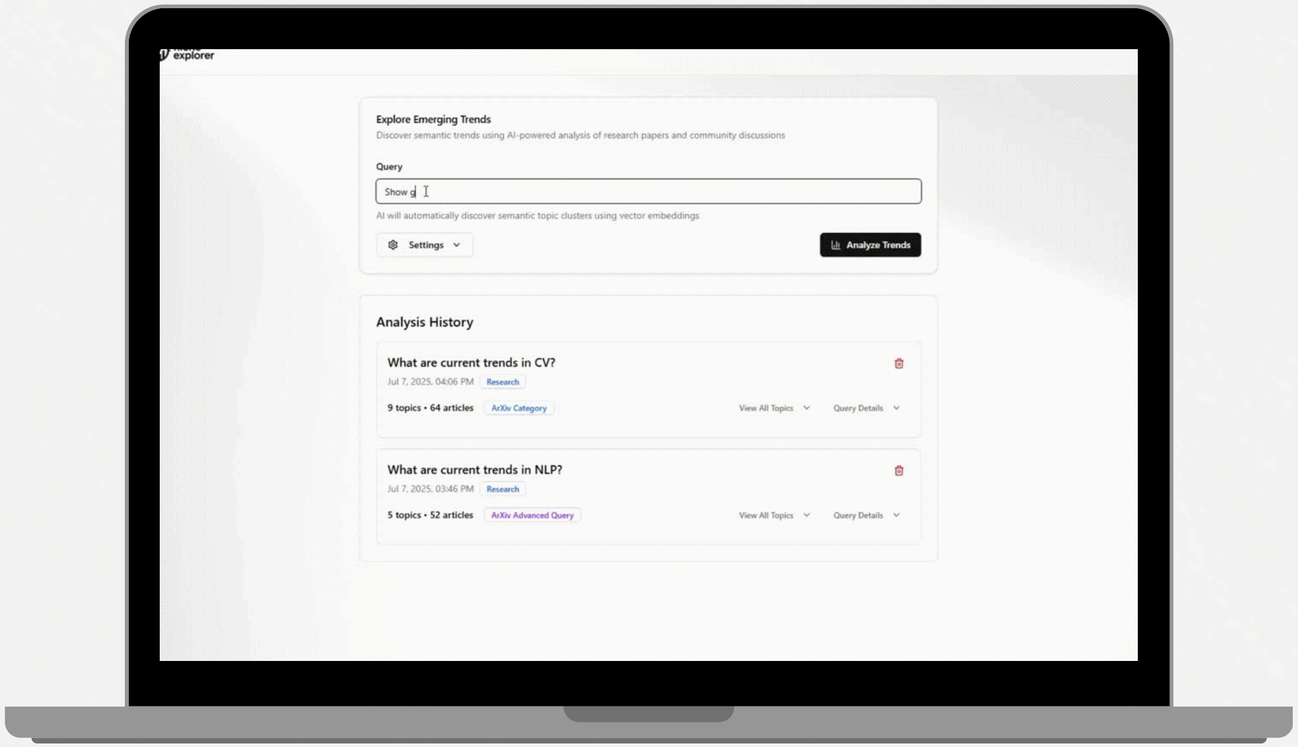
Niche Explorer - Janik Jehkul, Manuel Wilhelm

NicheExplorer

Problem: Great ideas are buried under a mountain of data

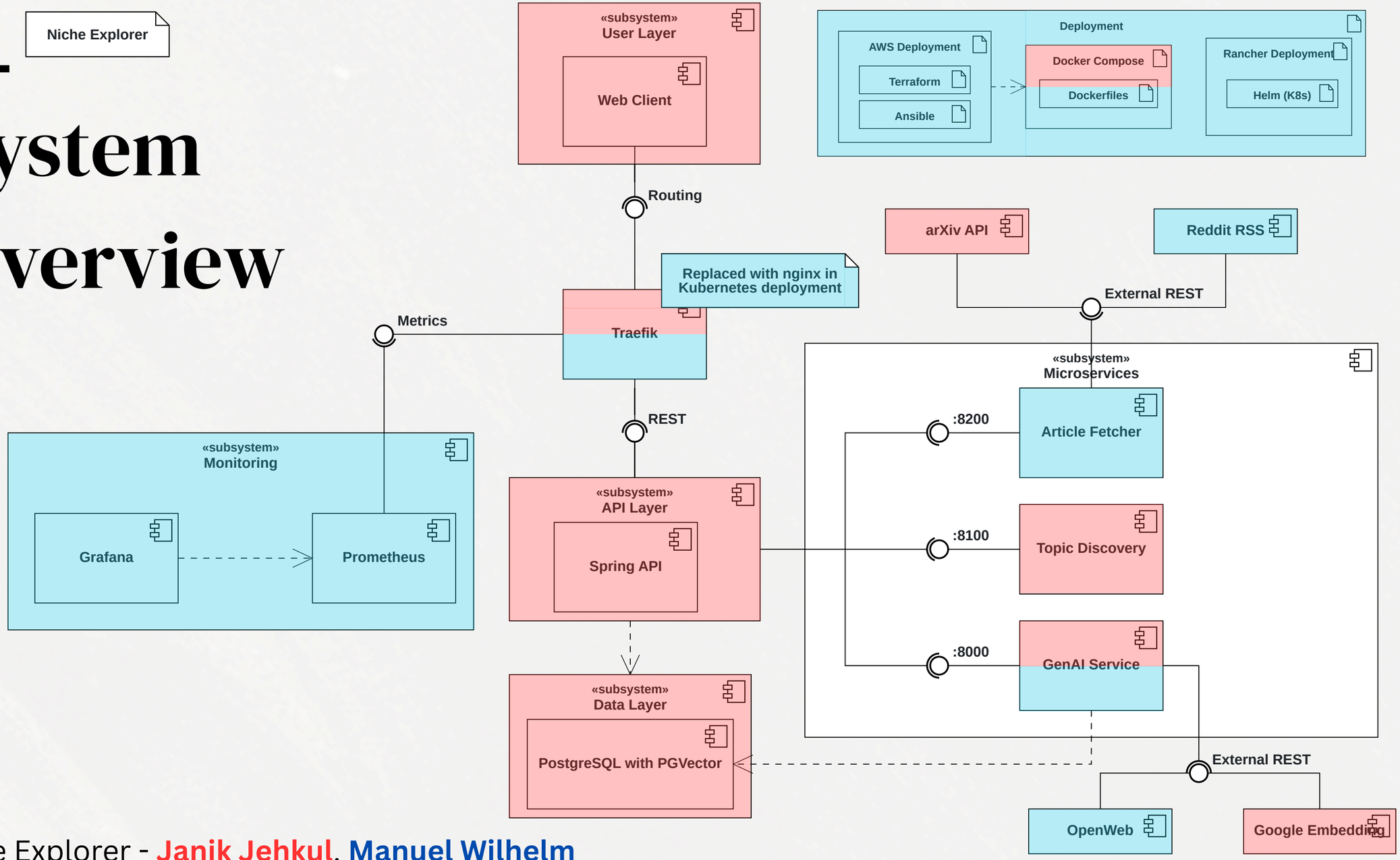
Solution:

- **You ask a question**
- NicheExplorer **fetches** latest related documents from **arXiv** or **reddit**
- **Identifies key trends** using topic modeling
- Returns an **interactive list with trends** you can explore



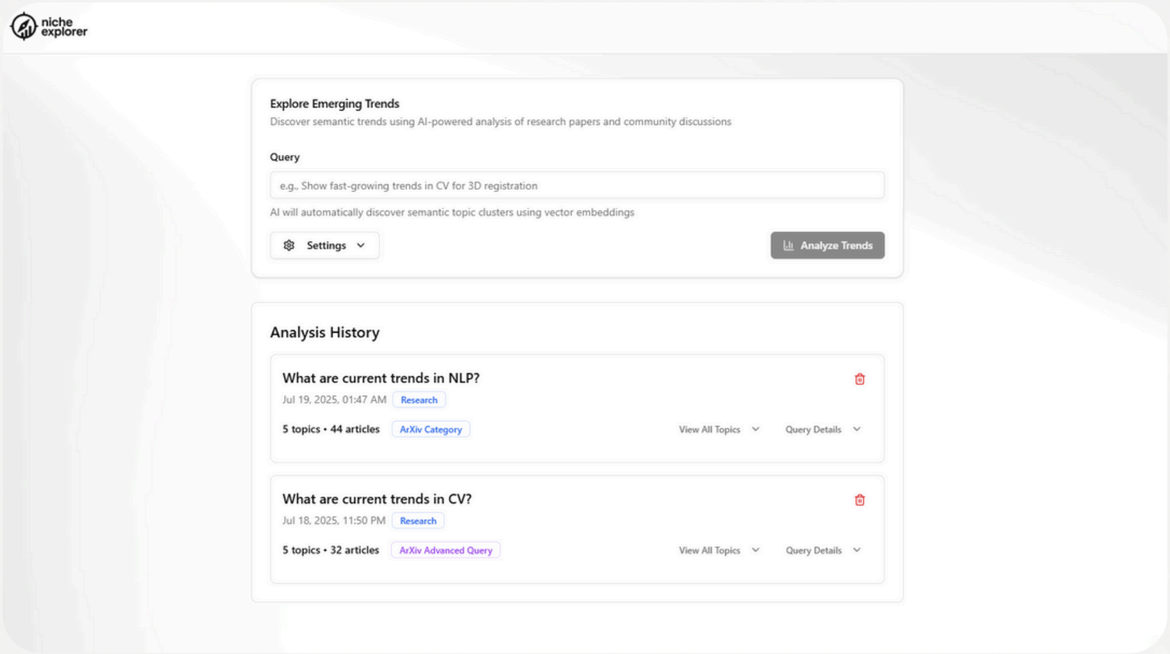
Niche Explorer

System Overview



Client - Frontend

- **Core stack:** React, Vite & TypeScript
- **UI Framework:** Tailwind CSS paired with shadcn components
- **API Contract:** Type-safe backend communication via OpenAPI-generated client



Server - Spring API

- **Core stack:** Spring Boot & Java, built with Gradle
- **Layered Architecture**:
  - **Controller**: Validate & route HTTP requests
  - **Services**:
    - **AnalysisService**: Core logic & PostgreSQL persistence
    - **AnalysisOrchestrationService**: Microservice coordination
  - **Config:** Centralized client configuration

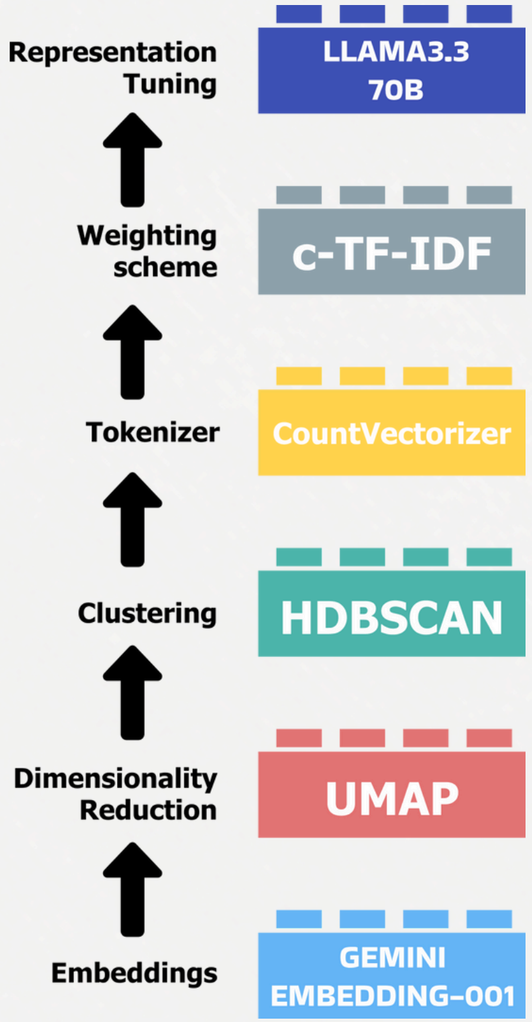
Python - Article Fetcher & Gen AI

- **Article Fetcher:** Python, FastAPI
  - Fetching ArXiv articles from the ArXiv API
  - Fetching Reddit posts from the Reddit RSS feed

- **Generative AI**:
  - **Text Generation**: Turns a prompt into structure text using OpenWebUI or Gemini
  - **Embedding**: Generates and fetches Gemini embeddings
  - **Classfication**: Predicts arXiv or reddit category & generates query

Python - Topic Discovery

- **Core stack:** Python, FastAPI
- **ML-Pipeline**:
  - Get document embeddings from PGVector
  - Cluster articles with BERTopic
  - Label/summarize each topic
- **Output**: Ranked list of topics with title, description relevance



Based on: maartengr.github.io/BERTopic

Monitoring

Prometheus (Metrics, Alerts)

- **Core metrics:**
  - Service Availability
  - Number of Calls to External Providers
- Core **Alerts**:
  - Service Unavailable
  - Calls Approaching Provider Rate Limit
  - Calls Exceeding Provider Rate Limit

Grafana (Visualization)



Deployment

Local - Docker Compose	Docker Compose Local Docker Compose Overwrite
Rancher - K8s Deployment	Helm: Monitoring Deplyoment Application Deployment
Cloud - AWS Deployment	(Terraform) Ansible Docker Compose AWS Docker Compose Overwrite

Dropped out Teammember

- **When:** Midway through the project
- **Contribution:** No real contribution to the project
  - Lead to more work (major refactorings)
- **Missing parts of the project due to that reason**:
  - Query Responses in frontend will just be empty on errors / when finding nothing
  - Smoother project startup (2-5 minutes wait time before everything running)
  - More test (such as client tests)
  - Credentials management (Keycloak)

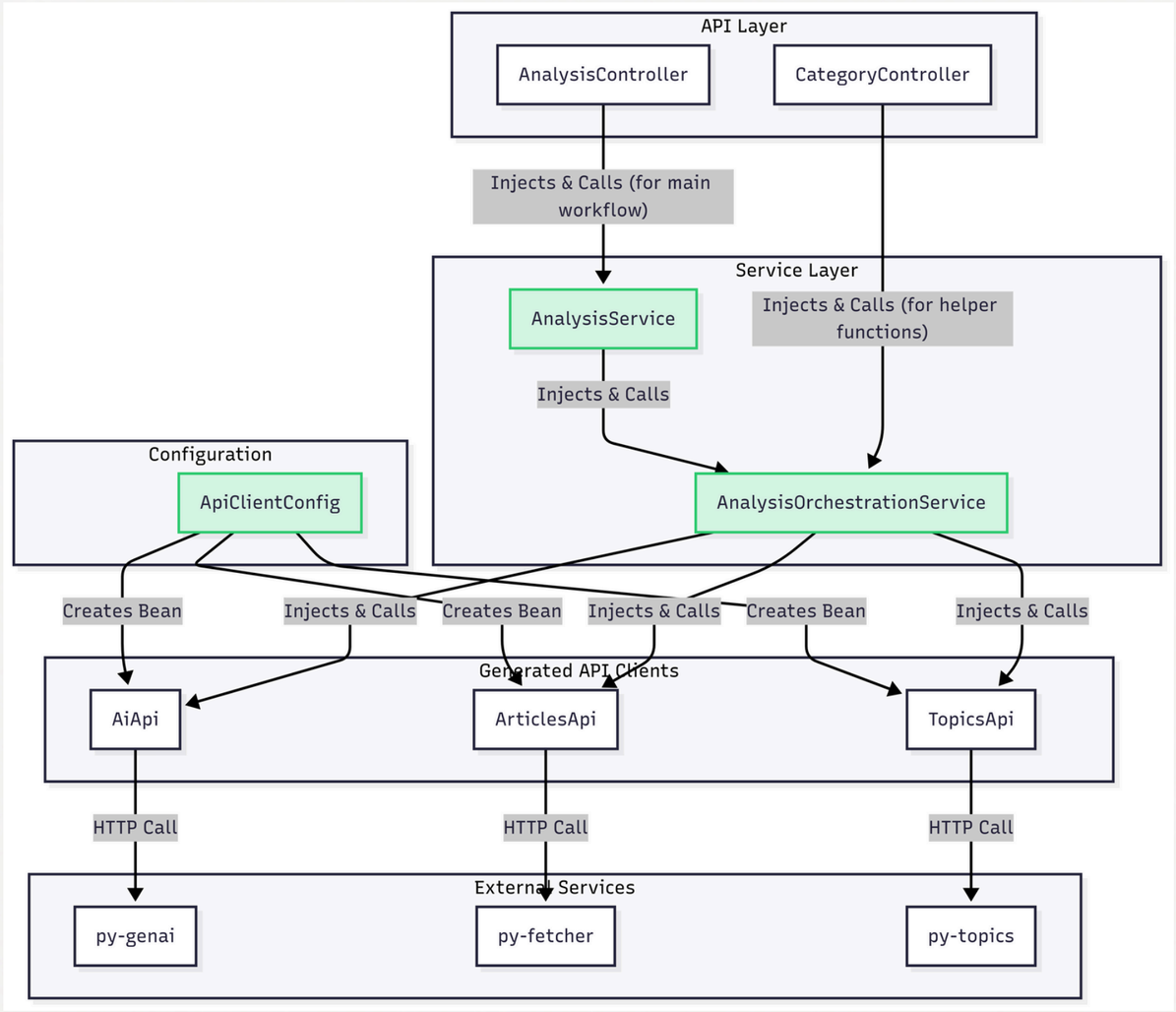
niche explorer

THANK YOU FOR LISTENING

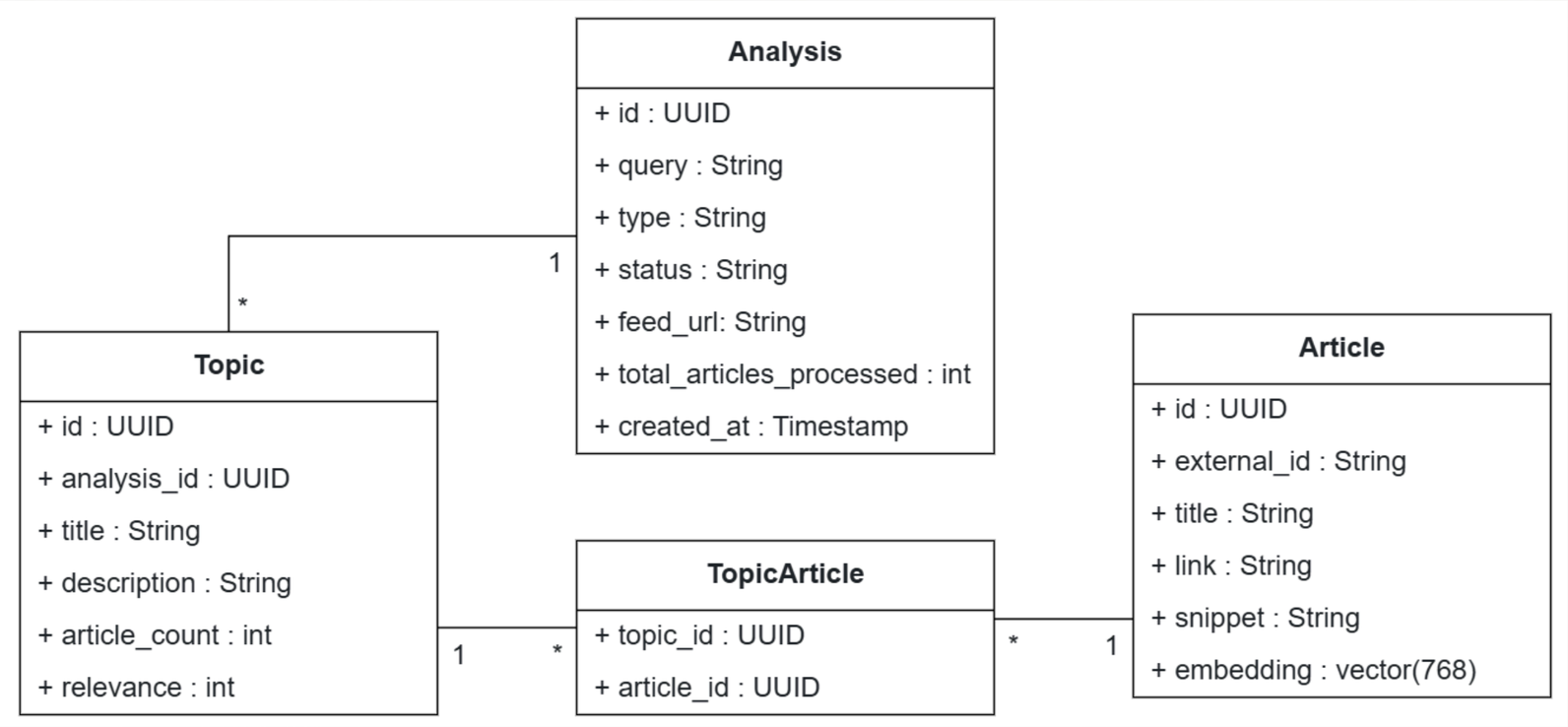
QUESTIONS?

Niche Explorer - Janik Jehkul, Manuel Wilhelm

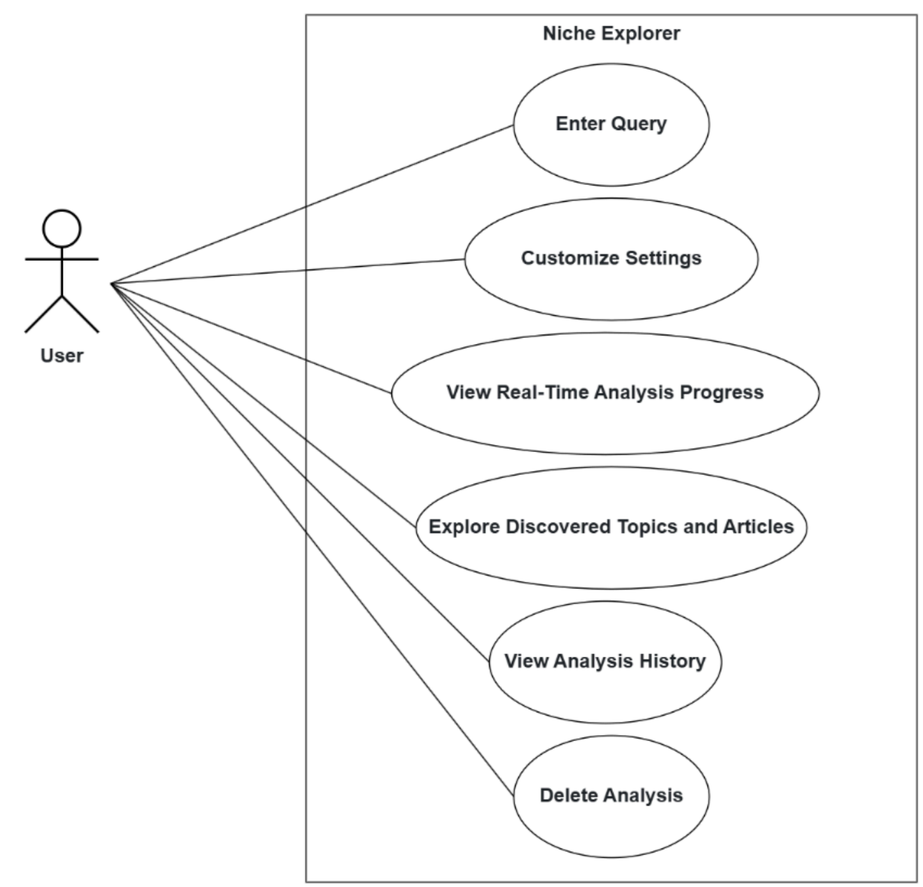
Appendix - Java



Appendix - Class Diagram



Appendix - Use Case Diagram



Appendix - Helpful links

Canva Presentation

SwaggerUI

Video demo

GitHub